Convergence Tests

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: The ratio and comparison tests**

Use the ratio test to determine whether $\sum_{n=1}^{\infty} \frac{n}{2^n}$ converges.

$\lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| = \lim_{n\to\infty} \frac{n+1}{2n} = \frac{1}{2} < 1$: converges.

2 **[Foundational]** • **Target: The ratio and comparison tests**

Use the ratio test on $\sum_{n=1}^{\infty} \frac{n!}{n^n}$.

$\frac{a_{n+1}}{a_n} = \frac{n^n}{(n+1)^n} = \left(\frac{n}{n+1}\right)^n \to \frac{1}{e} < 1$: converges.

3 **[Foundational]** • **Target: The ratio and comparison tests**

Use the comparison test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n^2+3}$ converges, by comparing to $\sum \frac{1}{n^2}$.

$0 < \frac{1}{n^2+3} < \frac{1}{n^2}$ for all $n \geq 1$, and $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$), so by comparison the given series converges.

4 **[Foundational]** • **Target: Alternating series and the p-series test**

Determine whether each p-series converges or diverges:

a $\sum_{n=1}^{\infty} \frac{1}{n^3}$ Converges ($p = 3 > 1$).

b $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$ Diverges ($p = \frac{1}{2} \leq 1$).

(a) Converges ($p = 3 > 1$).; (b) Diverges ($p = \frac{1}{2} \leq 1$).

5 **[Foundational]** • **Target: Alternating series and the p-series test**

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ converges, and classify it as absolutely convergent, conditionally convergent, or divergent.

The terms $\frac{1}{n}$ decrease to 0, so by the alternating series test the series converges. But $\sum \frac{1}{n}$ (absolute values) diverges — so the series is conditionally convergent.

6 **[Foundational]** • **Target: The integral test**

Use the integral test to determine whether $\sum_{n=1}^{\infty} \frac{1}{n \ln n}$ (starting at $n = 2$) converges or diverges, using $\int_2^{\infty} \frac{1}{x \ln x}\,dx$.

Let $u = \ln x$: $\int \frac{1}{x \ln x} dx = \ln|\ln x| + C$. $\int_2^{\infty} \frac{dx}{x \ln x} = \lim_{b\to\infty} [\ln(\ln x)]_2^b = \infty$: the integral diverges, so by the integral test the series diverges.

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7 [Foundational] • Target: The integral test

Use the integral test to confirm that $\sum_{n=1}^{\infty} \frac{1}{n^2}$ converges, using $\int_1^{\infty} \frac{1}{x^2}\,dx$.

$\int_1^{\infty} \frac{1}{x^2}dx = \lim_{b\to\infty} \left[-\frac{1}{x}\right]_1^b = 1$, a finite value, so by the integral test the series converges.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Choosing the right test

For each series, name the most efficient test to apply (ratio, comparison, p-series, or alternating series), without fully evaluating:

a $\sum \frac{2^n}{n!}$ Ratio test (factorial in denominator is a strong signal).

b $\sum \frac{(-1)^n}{\sqrt{n}}$ Alternating series test (terms decrease to 0).

c $\sum \frac{1}{n^4}$ p-series test directly ($p = 4 > 1$: converges).

(a) Ratio test (factorial in denominator is a strong signal).; (b) Alternating series test (terms decrease to 0).; (c) p-series test directly ($p = 4 > 1$: converges).

9 [Intermediate] • Target: The limit comparison test

Use the limit comparison test to determine whether $\sum_{n=1}^{\infty} \frac{n+1}{n^3+2}$ converges, comparing to $\sum \frac{1}{n^2}$.

$\lim_{n\to\infty} \frac{(n+1)/(n^3+2)}{1/n^2} = \lim_{n\to\infty} \frac{n^3+n^2}{n^3+2} = 1$, a finite positive value. Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

10 [Intermediate] • Target: The limit comparison test

Use the limit comparison test on $\sum_{n=1}^{\infty} \frac{3n^2-1}{n^4+2n}$, comparing to $\sum \frac{1}{n^2}$.

$\lim_{n\to\infty} \frac{(3n^2-1)/(n^4+2n)}{1/n^2} = \lim_{n\to\infty} \frac{3n^4-n^2}{n^4+2n} = 3$, finite and positive. Since $\sum \frac{1}{n^2}$ converges, so does the given series.

11 [Intermediate] • Target: The root test

Use the root test to determine whether $\sum_{n=1}^{\infty} \left(\frac{n}{2n+1}\right)^n$ converges.

$\lim_{n\to\infty} \sqrt[n]{\left|\left(\frac{n}{2n+1}\right)^n\right|} = \lim_{n\to\infty} \frac{n}{2n+1} = \frac{1}{2} < 1$: converges.

12 [Intermediate] • Target: The root test

Use the root test on $\sum_{n=1}^{\infty} \frac{3^n}{n^n}$.

$\lim_{n\to\infty} \sqrt[n]{\frac{3^n}{n^n}} = \lim_{n\to\infty} \frac{3}{n} = 0 < 1$: converges.

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13 [Intermediate] • Target: Absolute versus conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2}$ is absolutely convergent, conditionally convergent, or divergent.

Taking absolute values: $\sum \frac{1}{n^2}$ is a convergent p-series ($p = 2 > 1$). Since the series of absolute values converges, the original series is absolutely convergent.

14 [Intermediate] • Target: Absolute versus conditional convergence

Determine whether $\sum_{n=1}^{\infty} \frac{(-1)^n}{\sqrt{n}}$ is absolutely convergent, conditionally convergent, or divergent.

Absolute values give $\sum \frac{1}{\sqrt{n}}$, a divergent p-series ($p = \frac{1}{2} \leq 1$). But the terms $\frac{1}{\sqrt{n}}$ decrease to 0, so by the alternating series test the original series converges. It is conditionally convergent.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Estimating the sum of an alternating series • Cross-Lesson: 7.01 Integration by Substitution

The alternating series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ converges by the alternating series test. Use the fact that the error after n terms is bounded by the first omitted term to estimate how many terms are needed for the partial sum to be within 0.001 of the true sum.

Error bound: $|S - S_n| \leq a_{n+1} = \frac{1}{(n+1)^2}$. Need $\frac{1}{(n+1)^2} < 0.001$: $(n+1)^2 > 1000$, so $n + 1 > 31.6$, meaning $n \geq 32$ terms suffice.

16 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 7.02 Integration by Parts

Determine whether $\sum_{n=1}^{\infty} \frac{n^2}{3^n}$ converges, using the ratio test.

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{(n+1)^2}{3^{n+1}} \cdot \frac{3^n}{n^2} = \frac{1}{3}\left(\frac{n+1}{n}\right)^2 \to \frac{1}{3} < 1$: converges.

17 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 8.01 Area Between Curves

Determine whether $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$ converges, using the integral test with $u = \ln x$.

$\int_2^{\infty} \frac{1}{x(\ln x)^2}\,dx$: with $u = \ln x$, $du = \frac{1}{x}dx$: $\int \frac{1}{u^2}\,du = -\frac{1}{u} + C$. Evaluating: $\lim_{b\to\infty}\left[-\frac{1}{\ln x}\right]_2^b = 0 - \left(-\frac{1}{\ln 2}\right) = \frac{1}{\ln 2}$, a finite value: the integral converges, so by the integral test the series converges.

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18 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 8.02 Volumes of Revolution

A student claims that since $\lim_{n\to\infty} \frac{1}{n} = 0$, the series $\sum_{n=1}^{\infty} \frac{1}{n}$ must converge. Explain what is wrong with this reasoning, and state the correct necessary (but not sufficient) condition that the nth-term test actually provides, using the harmonic series as a counterexample.

The reasoning confuses a necessary condition with a sufficient one. The nth-term test only says: IF $\lim_{n\to\infty} a_n \neq 0$, THEN the series diverges. It says nothing when the limit IS 0 — that case is inconclusive. The harmonic series $\sum \frac{1}{n}$ is the standard counterexample: its terms go to 0, yet the series diverges (proven via the integral test or the comparison/grouping argument), showing that $\lim a_n = 0$ alone never guarantees convergence.

19 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 8.03 Arc Length and Improper Integrals

For the series $\sum_{n=1}^{\infty} \frac{n^3+2n}{2n^5-n+1}$, first find the nth-term limit to check the necessary condition, then use the limit comparison test with $\sum \frac{1}{n^2}$ to determine whether the series converges.

nth-term limit: $\lim_{n\to\infty} \frac{n^3+2n}{2n^5-n+1} = 0$ (inconclusive by itself). Limit comparison with $\frac{1}{n^2}$: $\lim_{n\to\infty} \frac{(n^3+2n)/(2n^5-n+1)}{1/n^2} = \lim_{n\to\infty} \frac{n^5+2n^3}{2n^5-n+1} = \frac{1}{2}$, finite and positive. Since $\sum \frac{1}{n^2}$ converges ($p = 2 > 1$), by the limit comparison test the given series also converges.

20 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 9.01 Separable Differential Equations

A quality-control engineer models cumulative defect probability using $\sum_{n=1}^{\infty} \frac{1}{n \cdot 2^n}$. Use the ratio test to confirm this series converges, and briefly explain (in one sentence) why a convergent bound like this is useful for guaranteeing that a manufacturing process's total defect risk stays finite over infinitely many production runs.

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{n \cdot 2^n}{(n+1) \cdot 2^{n+1}} = \frac{n}{2(n+1)} \to \frac{1}{2} < 1$: converges. Practically, a convergent bound guarantees the cumulative defect probability approaches a finite ceiling rather than growing without bound as more production runs occur, which is essential for meaningful long-run risk assessment.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Integral Test and p-Series

For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$ ✓

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

By the Integral Test, $\int_1^{\infty} x^{-p}\,dx$ converges if and only if $p > 1$. When $p \leq 1$, the p-series diverges.

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22 [Foundational] • Target: Integral Test and p-Series

Use the Integral Test to determine the convergence of $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$.

a. **Diverges ✓**

b. Converges to 1

c. Converges to $\ln 2$

d. Inconclusive

$\int_2^{\infty} \frac{1}{x \ln x}\,dx = [\ln(\ln x)]_2^{\infty} = \infty$. Since the integral diverges, the series diverges.

23 [Foundational] • Target: Integral Test and p-Series

Which conditions must $f(x)$ satisfy to apply the Integral Test to $\sum a_n$ where $a_n = f(n)$?

a. f **must be continuous, positive, and decreasing on** $[1, \infty)$ **✓**

b. f must be differentiable and alternating

c. f must be bounded and periodic

d. $f'(x) > 0$ everywhere

The Integral Test requires $f(x)$ to be continuous, positive, and eventually decreasing for all $x \geq 1$.

24 [Foundational] • Target: Comparison Tests (Direct and Limit)

Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3 + 5}$.

a. **Converges by Direct Comparison with** $\sum \frac{1}{n^3}$ **✓**

b. Diverges by comparison with $1/n$

c. Inconclusive

d. Diverges by the Divergence Test

For $n \geq 1$, $\frac{1}{n^3 + 5} < \frac{1}{n^3}$. Since $\sum 1/n^3$ is a convergent p-series ($p = 3 > 1$), the series converges.

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25 [Foundational] • Target: Comparison Tests (Direct and Limit)

Use the Limit Comparison Test with $b_n = 1/n$ to test $\sum_{n=1}^{\infty} \frac{2n+1}{n^2-3}$.

a. **Diverges because $L = 2 > 0$ and $\sum 1/n$ diverges** ✓

b. Converges because $L = 0$

c. Converges because $L = 2$

d. Inconclusive

$L = \lim_{n\to\infty} \frac{(2n+1)/(n^2-3)}{1/n} = \lim_{n\to\infty} \frac{2n^2+n}{n^2-3} = 2$. Since $0 < L < \infty$, both series share the same behavior. Thus it diverges.

26 [Foundational] • Target: Comparison Tests (Direct and Limit)

Which comparison series b_n is best suited for testing $\sum_{n=1}^{\infty} \frac{\sqrt{n}}{n^2+4n}$ via LCT?

a. $b_n = \frac{1}{n^{3/2}}$ ✓

b. $b_n = \frac{1}{n^2}$

c. $b_n = \frac{1}{n}$

d. $b_n = \frac{1}{\sqrt{n}}$

The dominant terms are $\frac{\sqrt{n}}{n^2} = \frac{n^{1/2}}{n^2} = \frac{1}{n^{3/2}}$. Since $p = 3/2 > 1$, it is a convergent p-series.

27 [Foundational] • Target: Alternating Series and Error Bounds

What are the two conditions of the Alternating Series Test (AST) for $\sum (-1)^n b_n$ ($b_n > 0$)?

a. $b_{n+1} \le b_n$ **for all** n **and** $\lim_{n\to\infty} b_n = 0$ ✓

b. $\sum b_n$ converges and $b_n > 0$

c. $\lim_{n\to\infty} b_n = 1$ and b_n is decreasing

d. b_n is geometric

The Leibniz criteria require that the terms are positive, monotonically decreasing ($b_{n+1} \le b_n$), and have limit 0.

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28 [Foundational] • Target: Alternating Series and Error Bounds

Classify $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$.

a. **Conditionally convergent ✓**

b. Absolutely convergent

c. Divergent

d. Oscillating divergent

The alternating harmonic series converges by AST. However, the series of absolute values $\sum 1/n$ is the divergent harmonic series. Thus it is conditionally convergent.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Alternating Series and Error Bounds

If $S = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^4}$ is approximated by S_3, what is the maximum error by the Alternating Series Estimation Theorem?

a. $|R_3| \le \frac{1}{4^4} = \frac{1}{256}$ ✓

b. $|R_3| \le \frac{1}{3^4} = \frac{1}{81}$

c. $|R_3| \le \frac{1}{5^4}$

d. $|R_3| \le \frac{1}{16}$

By the Alternating Series Estimation Theorem, $|R_n| \le b_{n+1}$. For $n = 3$, $|R_3| \le b_4 = \frac{1}{4^4} = \frac{1}{256}$.

30 [Intermediate] • Target: Ratio and Root Tests

State the conclusion of the Ratio Test when $L = \lim_{n\to\infty} \left|\frac{a_{n+1}}{a_n}\right| < 1$.

a. **The series converges absolutely ✓**

b. The series converges conditionally

c. The series diverges

d. The test is inconclusive

If $L < 1$, the series converges absolutely. If $L > 1$, it diverges. If $L = 1$, the Ratio Test is inconclusive.

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31 [Intermediate] • Target: Ratio and Root Tests

Apply the Ratio Test to $\sum_{n=1}^{\infty} \frac{3^n}{n!}$.

a. $L = 0 < 1 \Rightarrow$ **Converges absolutely** ✓

b. $L = 3 > 1 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = \infty \Rightarrow$ Diverges

$\left|\frac{a_{n+1}}{a_n}\right| = \frac{3^{n+1}}{(n+1)!}\frac{n!}{3^n} = \frac{3}{n+1} \to 0$ as $n \to \infty$. Since $L = 0 < 1$, the series converges absolutely.

32 [Intermediate] • Target: Ratio and Root Tests

Use the Root Test to determine the convergence of $\sum_{n=1}^{\infty} \left(\frac{2n+1}{3n-1}\right)^n$.

a. $L = 2/3 < 1 \Rightarrow$ **Converges absolutely** ✓

b. $L = 2/3 \Rightarrow$ Diverges

c. $L = 1 \Rightarrow$ Inconclusive

d. $L = 0$

$\lim_{n\to\infty} \sqrt[n]{|a_n|} = \lim_{n\to\infty} \frac{2n+1}{3n-1} = \frac{2}{3}$. Since $L = 2/3 < 1$, the series converges absolutely.

33 [Intermediate] • Target: Alternating Series and Error Bounds

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 10001

b. 9999

c. 20000

d. 10000 ✓

The error after summing up to $n = 10000$ is at most the next term, $\frac{1}{10001} < \frac{1}{10000}$, while stopping one term earlier leaves a bound $\frac{1}{10000} \geq \frac{1}{10000}$. So 10000 terms are needed.

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34 [Intermediate] • Target: Ratio and Root Tests

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 101

b. 100 ✓

c. 99

d. 10000

The error after summing up to $n = 100$ is at most the next term, $\frac{1}{10201} < \frac{1}{10000}$, while stopping one term earlier leaves a bound $\frac{1}{10000} \geq \frac{1}{10000}$. So 100 terms are needed.

35 [Intermediate] • Target: The ratio and comparison tests

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ needed to guarantee an error less than $\frac{1}{1000}$?

a. 1001

b. 999

c. 1000 ✓

d. 2000

The error after summing up to $n = 1000$ is at most the next term, $\frac{1}{1001} < \frac{1}{1000}$, while stopping one term earlier leaves a bound $\frac{1}{1000} \geq \frac{1}{1000}$. So 1000 terms are needed.

36 [Intermediate] • Target: Alternating series and the p-series test

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^{n}}{n!}$ needed to guarantee an error less than $\frac{1}{10000}$?

a. 8 ✓

b. 9

c. 7

d. 10000

The error after summing up to $n = 7$ is at most the next term, $\frac{1}{40320} < \frac{1}{10000}$, while stopping one term earlier leaves a bound $\frac{1}{5040} \geq \frac{1}{10000}$. So 8 terms are needed.

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37 [Intermediate] • Target: The integral test

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ needed to guarantee an error less than $\frac{1}{1000}$?

a. 31 ✓

b. 32

c. 30

d. 1000

The error after summing up to $n = 31$ is at most the next term, $\frac{1}{1024} < \frac{1}{1000}$, while stopping one term earlier leaves a bound $\frac{1}{961} \geq \frac{1}{1000}$. So 31 terms are needed.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Choosing the right test • Cross-Lesson: 8.02 Volumes of Revolution

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^{n}}{n!}$ needed to guarantee an error less than $\frac{1}{1000}$?

a. 8

b. 6

c. 7 ✓

d. 1000

The error after summing up to $n = 6$ is at most the next term, $\frac{1}{5040} < \frac{1}{1000}$, while stopping one term earlier leaves a bound $\frac{1}{720} \geq \frac{1}{1000}$. So 7 terms are needed.

39 [Challenging] • Target: The limit comparison test • Cross-Lesson: 8.03 Arc Length and Improper Integrals

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n^2}$ needed to guarantee an error less than $\frac{1}{100}$?

a. 11

b. 9

c. 100

d. 10 ✓

The error after summing up to $n = 10$ is at most the next term, $\frac{1}{121} < \frac{1}{100}$, while stopping one term earlier leaves a bound $\frac{1}{100} \geq \frac{1}{100}$. So 10 terms are needed.

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40 [Challenging] • Target: The root test • Cross-Lesson: 9.01 Separable Differential Equations

By the alternating series estimation theorem, what is the fewest number of terms of $\sum_{n=0}^{\infty} \frac{(-1)^n}{n!}$ needed to guarantee an error less than $\frac{1}{100}$?

a. 6

b. 4

c. 100

d. 5 ✓

The error after summing up to $n = 4$ is at most the next term, $\frac{1}{120} < \frac{1}{100}$, while stopping one term earlier leaves a bound $\frac{1}{24} \geq \frac{1}{100}$. So 5 terms are needed.

41 [Challenging] • Target: Absolute versus conditional convergence • Cross-Lesson: 1.01 Precalculus Review for Calculus

Series $\sum_{n=1}^{\infty} (-1)^{n-1}/\sqrt{n}$ converges by alternating series test. Is it absolutely convergent?

a. Yes, $p = 1/2 < 1$ diverges

b. **No, $\sum 1/\sqrt{n}$ diverges ($p = 1/2 \leq 1$)** ✓

c. Yes, $\sum 1/\sqrt{n}$ converges

d. Cannot determine

$p = 1/2 \leq 1 \Rightarrow \sum 1/\sqrt{n}$ diverges: only conditionally convergent.

42 [Challenging] • Target: Estimating the sum of an alternating series • Cross-Lesson: 10.01 Calculus with Parametric and Polar Curves

Integral test convergence of $\sum_{n=1}^{\infty} 1/n(\ln n)^2$ (from $n = 2$) gives sum approximately:

a. **converges** ✓

b. diverges

c. equals e

d. equals π^2

$\int_2^{\infty} dx/(x(\ln x)^2) = 1/\ln 2 < \infty \Rightarrow$ converges.

43 [Challenging] • Target: A combined convergence-test challenge • Cross-Lesson: 11.01 Sequences and Series

Which test determines $\sum n^2/e^n$? Answer: by ratio test $L =$

a. e

b. 1

c. $1/e < 1$ ✓

d. e^2

$L = \lim (n+1)^2/e^{n+1} \cdot e^n/n^2 = 1/e < 1 \Rightarrow$ converges.

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44 [Challenging] • Target: A multi-step convergence investigation • Cross-Lesson: 12.02 Power Series and Taylor Series

Series $\sum_{n=1}^{\infty} n \cdot (1/2)^n$: ratio test gives $L =$

a. 1

b. 2

c. 1/2 ✓

d. $n/2$

$|a_{n+1}/a_n| = (n+1)/(2n) \to 1/2 < 1 \Rightarrow$ converges.

45 [Challenging] • Target: Integral Test and p-Series • Cross-Lesson: 13.01 Vectors and Vector-Valued Functions

The ratio test for $\sum_{n=0}^{\infty} x^n/n!$: $|a_{n+1}/a_n| = |x|/(n+1) \to$

a. $|x|$

b. 1

c. 0 ✓

d. ∞

$|x|/(n+1) \to 0 < 1$ for all $x \Rightarrow R = \infty$.

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